# 1. Odd, mixed and isogeny PKM towers under F4, and the algebraic relation oracle against enumeration

EXP-PKM-ODD-001 · 2026-11-14 · isogeny-labs/crypto PR #4091 · protocol `PROTOCOL.md` (frozen in `fff5b8f` before any study row; Amendment 1 run order, Amendment 2 analysis bug fix) · rows `runs/` · analysis `analyze.py` → `results.interim.json`. **Interim report:** every `m = 2` and part B cell is complete. The three `m = 3` part A cells and three `m = 3` part B cells with `S ≥ 27` (Amendment 1 tail) are still running, so the `m = 3` verdicts below are read on `S ≤ 25` only. Counted metrics, unisolated local host, 4 threads [pcc:unverified].

## 1.1. Results

| question | verdict (frozen rule) | what was measured |
|---|---|---|
| Validity (C1, C2, A/A) | **valid** | 256 part A rows: F4′s verdict equals the exhaustive one on all of them. 16 planted rows check. A/A bit-identical (2,537,218,131 ×3). |
| Positive control | **detects one bit** | Doubling `\|V\|` on the degree-2 tower multiplies F4 work by 8.7–13.9 at every step. |
| Degree 3 vs degree 2 (`m = 2`) | **gain**, constant | `r = 0.24, 0.29, 0.30` at `S = 27, 81, 243`. Null: gain. Replicated at `P2`. |
| Degree 5 vs degree 2 (`m = 2`) | **gain** | `r = 0.16, 0.12` at `S = 25, 125`. Null: gain. Replicated at `P2`. |
| Mixed 2,3,… (`m = 2`) | **no material effect** | `r = 0.55, 0.55, 0.35, 0.69` at `S = 36, 72, 216, 432`. Null: same. |
| Mixed 3,2,… (`m = 2`) | **gain**, growing by the rule | `r = 0.49, 0.48, 0.38` at `S = 36, 108, 216`. Null: gain. |
| Order 3,2 vs 2,3 at equal `S` | **no material effect** | `0.89` at `S = 36`, `1.07` at `S = 216`, same targets. |
| Isogeny vs Kummer, same degrees | **no material effect**, all three | I2/A2 `0.76–1.47`; I3/A3 `1.17–1.32`; I23/A23 `0.88–1.22`. Null: same. |
| `m = 3` (interim, `S ≤ 25`) | A23 **no material effect**; A3 **no material effect** (tower) vs **gain** (null) | A3 `0.54` at `S = 9` (the null sits just under 0.5). A5 and A32 have no completed in-range cell yet. |
| Part B validity (C7) | **valid** | 76/76 targets: the algebraic oracle returns exactly enumeration's relation set. |
| Part B: algebraic vs enumeration | **disadvantage** in every cell | `m = 2`: 213 ⟶ $1.8{\cdot}10^5$ (A2, `S = 16 → 256`), 220 ⟶ $5.2{\cdot}10^4$ (A3), 317 ⟶ $5.9{\cdot}10^3$ (I3). `m = 3`: 317 ⟶ $7.7{\cdot}10^5$ (A2, `S = 4 → 16`). Trend: widening (A2) or flat by the rule; never closing. |

**Growth.** The least-squares slope of $\log_2$ `field_ops` against $\log_2$ `S` at `m = 2` is 3.11–3.60 for every arm, Kummer and isogeny (descriptive, no interval). All three draws of each cell gave identical counts, so the F4 work here is set by the system's shape, not the target.

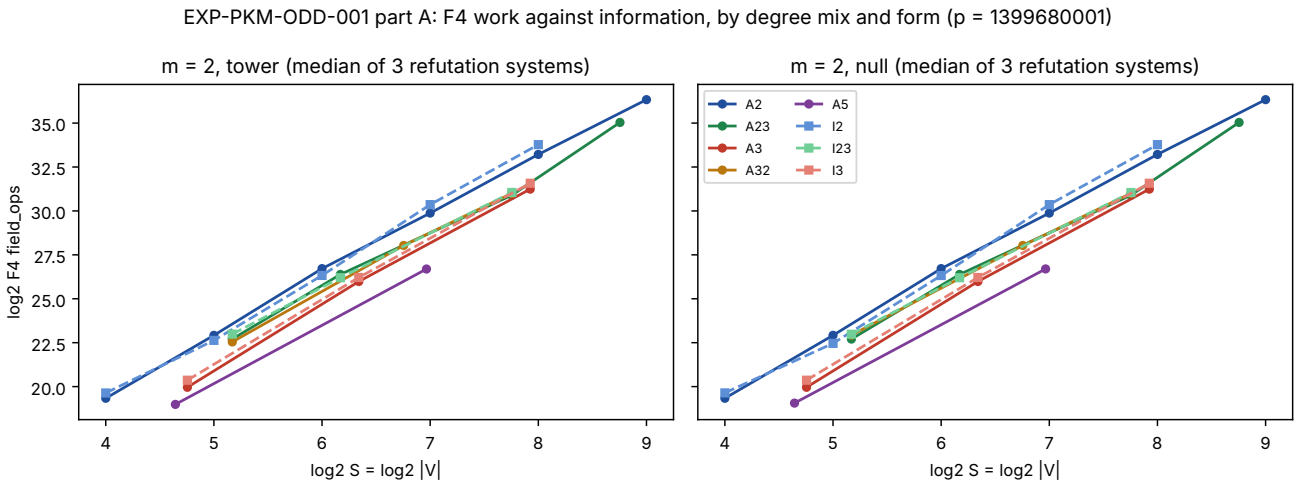



Figure 1: Part A, `m = 2`. F4 work against $\log_2 |V|$ for each degree mix and form, tower (left) and the null control (right). The lines are parallel: the degree mix moves the constant, not the slope.

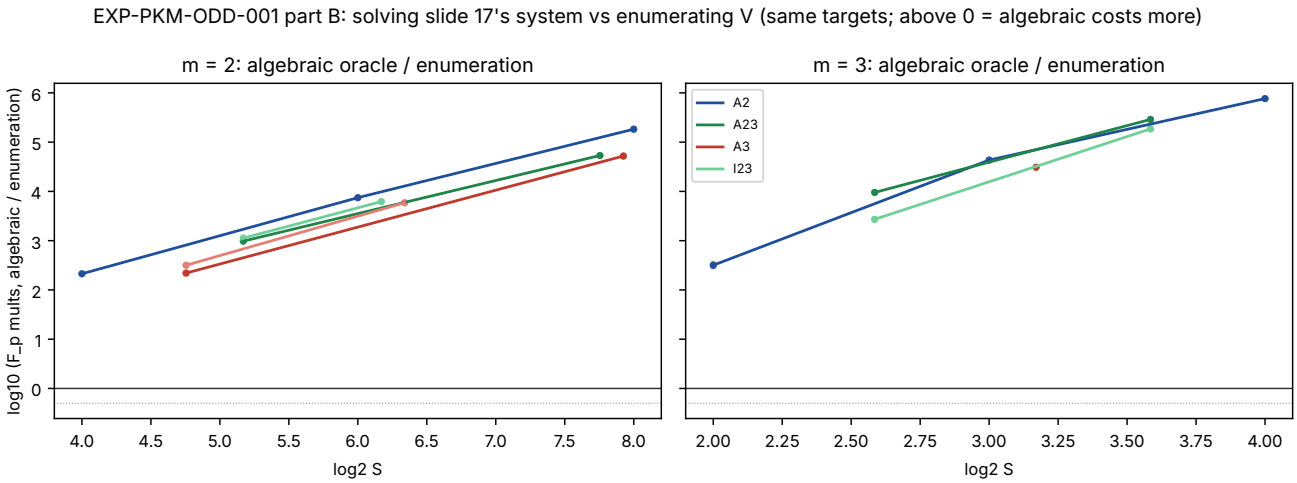



Figure 2: Part B. Algebraic oracle (F4 plus linear stage) over enumeration, in `F_p` multiplications, on the same targets (2 planted, 2 random per cell). Above 0 the algebraic oracle costs more. The dotted line is the 0.5 band.

## 1.2. What this says about the questions asked

- **Odd degrees.** Writing `V` with degree-3 or degree-5 steps cut F4's work by a constant factor of 3–8 at equal `|V|`. The prediction (no effect) is **falsified** for A3 and A5. The null control shows the same gain, so it belongs to the system's shape: fewer, higher-degree tower variables. No arm changed the growth rate, which is the barrier note's reading. A constant factor is engineering, not an exponent.
- **The auxiliary curve.** The isogeny form was never cheaper than the Kummer form beyond the band. Its F4 work was 0.76–1.47× Kummer's for the same degree sequence, and **the auxiliary-curve search was not charged**. The pre-registered "isogeny is not cheaper" prediction stands.
- **The slides' algebraic relation search.** Implemented, verified exact (76/76), and measured. On these sizes it costs between $2{\cdot}10^2$ and $8{\cdot}10^5$ times plain enumeration of the factor base, and the gap widens with `S` for the degree-2 tower at both `m = 2` and `m = 3`. The slides' hope that solving beats enumerating at `m ≥ 3` is not seen at `S ≤ 16`. The `S ≥ 27` cells are in the tail.

## 1.3. Scope and deviations

Kummer and isogeny towers over two primes near `2^30` and `2^31.6`; `m = 2` to $\log_2$ `S = 9`, `m = 3` small. Nothing here is about cryptographic sizes or Pollard rho, and no claim is made beyond the ladder.

- **Backend:** local, because there is no EKS route.
- **Checkpoint:** `CHECKPOINT_SKIP=no-s3-credentials`, recorded in the pcc ledger.
- **Model tiers:** the director/executor relay was not used, because it needs an S3 benchmark record.
- **Knowledge base:** not available. Searched with `git grep`.
- **Root finding:** the univariate root finding in part B is not counted, which favours the algebraic oracle.

Details are in `PROTOCOL.md` §7–8.